

Final Project Proposal

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Goal

- Our goal is to analyze a dataset focused on movie statistics and see how certain factors affect each other
 - What factors affect the score and gross profit of a movie

Dataset

- <https://github.com/danielgrijalva/movie-stats/blob/master/movies.csv>

Content

There are 6820 movies in the dataset (220 movies per year, 1986-2016). Each movie has the following attributes:

- budget: the budget of a movie. Some movies don't have this, so it appears as 0
- company: the production company
- country: country of origin
- director: the director
- genre: main genre of the movie.
- gross: revenue of the movie
- name: name of the movie
- rating: rating of the movie (R, PG, etc.)
- released: release date (YYYY-MM-DD)
- runtime: duration of the movie
- score: IMDb user rating
- votes: number of user votes
- star: main actor/actress
- writer: writer of the movie
- year: year of release

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	name	rating	genre	year	released	score	votes	director	writer	star	country	budget	gross	company	runtime
2	The Shining	R	Drama	1980	June 13, 1980	8.4	927000	Stanley Kubrick	Stephen King	Jack Nicholson	United Kingdom	19000000	46998772	Warner Bros.	146
3	The Blue Lagoon	R	Adventure	1980	July 2, 1980	5.8	65000	Randal Kleiser	Henry De Vere	Brooke Shields	United States	4500000	58853106	Columbia	104
4	Star Wars	PG	Action	1980	June 20, 1980	8.7	1200000	Irvin Kershner	Leigh Brackett	Mark Hamill	United States	18000000	5.38E+08	Lucasfilm	124
5	Airplane!	PG	Comedy	1980	July 2, 1980	7.7	221000	Jim Abrahams	Jim Abrahams	Robert Harris	United States	3500000	83453539	Paramount	88
6	Caddyshack	R	Comedy	1980	July 25, 1980	7.3	108000	Harold Ramis	Brian Doyle	Chevy Chase	United States	6000000	39846344	Orion Pictures	98
7	Friday the 13th	R	Horror	1980	May 9, 1980	6.4	123000	Sean S. Cunningham	Victor Mill	Betsy Palmer	United States	550000	39754601	Paramount	95
8	The Blues Brothers	R	Action	1980	June 20, 1980	7.9	188000	John Landis	Dan Aykroyd	John Belushi	United States	27000000	1.15E+08	Universal	133
9	Raging Bull	R	Biography	1980	December 19, 1980	8.2	330000	Martin Scorsese	Jake LaMotta	Robert De Niro	United States	18000000	23402427	Chartoff-Vincent	129
10	Superman	PG	Action	1980	June 19, 1980	6.8	101000	Richard Donner	Jerry Siegel	Gene Hackman	United States	54000000	1.08E+08	Doveman	127
11	The Long Walk Home	F	Biography	1980	May 16, 1980	7	10000	Walter Hill	Bill Bryden	David Carr	United States	10000000	15795189	United Artists	100
12	Any Which Way You Can	PG	Action	1980	December 19, 1980	6.1	18000	Buddy Van Stan	Stanford S. Jaffe	Clint Eastwood	United States	15000000	70687344	The Malpaso Company	116
13	The Gods of Greed	PG	Adventure	1980	October 24, 1980	7.3	54000	Jamie Uys	Jamie Uys	Nxaxha	South Africa	5000000	30031783	C.A.T. Film	109
14	Popeye	PG	Adventure	1980	December 19, 1980	5.3	30000	Robert Altman	Jules Feiffer	Robin Williams	United States	20000000	49823037	Paramount	114
15	Ordinary People	P	Drama	1980	September 19, 1980	7.7	49000	Robert Redford	Judith Gura	Donald Sutherland	United States	6000000	54766923	Paramount	124
16	Dressed to Kill	R	Crime	1980	July 25, 1980	7.1	37000	Brian De Palma	Brian De Palma	Michael Caine	United States	6500000	31899000	Filmways	104
17	Somewhere in Time	PG	Drama	1980	October 3, 1980	7.2	27000	Jean YVES Escoffier	Suzanne	Richard M. Crayton	United States	5100000	9709597	Rastar Pictures	103
18	Fame	R	Drama	1980	May 16, 1980	6.6	21000	Alan Parker	Christophe Yves	Eddie Bart	United States		21202829	Metro-Goldwyn-Mayer	134
19	9 to 5	PG	Comedy	1980	December 19, 1980	6.9	29000	Colin Higgins	Patricia Res	Jane Fonda	United States	10000000	1.03E+08	IPC Films	109
20	The Fog	R	Horror	1980	February 8, 1980	6.8	66000	John Carpenter	John Carpenter	Adrienne E. Barbeau	United States	1000000	21448782	AVCO Embassy	89
21	Stir Crazy	R	Comedy	1980	December 19, 1980	6.8	26000	Sidney Poitier	Bruce Jay	Gene Wilder	United States		1.01E+08	Columbia	111

Data We Need

- Movie genres
- Director
- Budget
- Score (IMDb User Rating)
- Gross Profit
- Country of Origin

Techniques

- KNN Regression Models
 - Predict how high a movie's score will be based on our specified factors
 - Predict the amount of revenue a movie may make based on our specified factors
- Collaborative Filtering
 - Based on one movie rating, find other similar movies whose ratings correlate with current movie